

Lab: Biological Boundaries -- Cells, Organs, and the Markov Blanket of Life

Objective

Investigate how biological systems at multiple scales (cells, organs, body systems) maintain boundaries that separate their internal states from the environment, and connect these boundaries to the concept of the Markov blanket.

Prerequisites

- Completed the Everyday Life section on Systems
- Read the module on Biological Boundaries
- Basic biology knowledge (cell structure, organ systems)

Part 1: The Cell as a System

Goal: Map the Markov blanket of a single cell.

Using your knowledge of cell biology, fill in the boundary diagram:

Component	Role in Markov Blanket	Examples
Internal states	Hidden states the environment cannot observe directly	
Sensory states (inputs)	How the cell receives information from outside	
Active states (outputs)	How the cell acts on its environment	
Blanket boundary	The physical structure forming the boundary	

Questions: 1. What is the cell membrane's role as a Markov blanket? 2. What happens to the cell if its membrane is destroyed? How does this relate to the Active Inference idea that a system ceases to exist without its boundary? 3. How do receptor proteins on the cell membrane function as "sensory states"?

Write your response here

Part 2: Organ System Boundaries

Goal: Identify Markov blankets at the organ-system level.

Choose two organ systems and map their boundaries:

Feature	Organ System 1: ____	Organ System 2: ____
What is "inside" this system?		
What are its sensory inputs?		
What are its active outputs?		
What does it regulate (preferred states)?		
What happens if the boundary fails?		

Write your response here

Part 3: Nested Boundaries

Goal: Explore how biological systems are nested inside each other.

Trace the nesting from smallest to largest:

- 1. **Organelle** (e.g., mitochondrion): What is its boundary? What does it regulate?
- 2. **Cell**: How does the cell boundary contain organelles while having its own blanket?
- 3. **Tissue**: How do cells organize into a tissue with a collective boundary?
- 4. **Organ**: How does the organ boundary emerge from tissues?
- 5. **Body**: How does skin function as the outermost Markov blanket?

Draw or describe this nesting structure. How does each level maintain its own homeostasis while contributing to the level above?

Write your response here

Part 4: Homeostasis Investigation

Goal: Research one specific homeostatic process and map it as a feedback loop.

Choose one: body temperature regulation, blood sugar regulation, or blood pH regulation.

Step	What Happens	Active Inference Term
Normal state (set point)		Preferred state
Disturbance occurs		Prediction error
Sensor detects change		Sensory state
Control center processes signal		Internal state update
Effector responds		Active state
Return to set point		Free energy minimized

Write your response here

Summary Table

Concept	Biological Example	Active Inference Connection
Markov Blanket	Cell membrane	Physical boundary with sensory and active components
Internal States	DNA, cytoplasm	Hidden from environment, only accessible through blanket
Homeostasis	Thermoregulation	Maintaining preferred states through action
Nested Systems	Organelle in cell in organ	Blankets within blankets